

Suleyman Kiani

Software Engineer · Fintech & Agentic Systems

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SUMMARY

Software engineer who builds the tooling a finance desk actually runs on: a production quoting engine matching TValue, the industry lease-math standard, to the penny; a multi-tenant booking-and-payments platform processing real client revenue; a self-hosted agentic platform with sandboxed workers and fail-closed gates. Equipment-finance experience at Mitsubishi HC Capital gives me domain fluency most engineers don't have. MEng in Computing & Software at McMaster, in distributed systems, microservices, and formal verification.

EDUCATION

McMaster University

Hamilton, ON

MEng, Computing & Software (Distributed Systems, Microservices & Applied ML)

Sept. 2025 – 2026

- Research with Dr. Farmer and Dr. Paige: TLA+ and SMT2 specifications for an LLM citation-faithfulness gating layer, verifying that a study companion's generated notes stay attributable to their sources.
- **A+** in Simple Type Theory and Microservices-Oriented Architectures.
- Built **PodcastHub** for the distributed-systems course: five event-driven microservices (Node/Express, Python/FastAPI) in six containers with a dedicated FFmpeg worker, choreographed over a RabbitMQ topic exchange with MinIO presigned-URL storage and a RabbitMQ-to-WebSocket notification bridge.

McMaster University

Hamilton, ON

BASc (Honours), Computer Science

Sept. 2018 – Nov. 2024

EXPERIENCE

Associate Account Manager (Internal Software Engineering)

Sept. 2025 – Present

Mitsubishi HC Capital Canada

Burlington, ON

- Rebuilt **13 abandoned Excel calculators** (30–45 sheets each) as one bilingual Next.js / React app covering **14 manufacturers**; reverse-engineered their lease and subsidy math, matched to the TValue schedules to the penny behind a **1,396-test** suite.
- Untangled three structurally different subsidy formulas, proved three regional dealer workbooks were re-skins of one template, and collapsed them into shared calculation engines; authored the architecture and handoff documentation that lets the company's IT org own and extend the tool.
- Built an AI document-validation service (LLM extraction + deterministic rules) for deal-funding review and carried it through the company's internal innovation program; on a **50-package evaluation suite** it cuts the **1–2 hour** first pass to about **a minute**, advisory only, on a replayable per-deal audit trail.
- Designed the collections risk scorecard the business approved in Aug 2026, now **in pilot**: it scores and ranks all **2,079 live delinquent contracts**, reordering **73%** of the top 100 against a plain balance sort. Derived the NSF, term, and industry multipliers from a **six-year** delinquency history, validated on a customer-grouped holdout with five-fold cross-validation, and built the scorer that reproduces every ranked score in plain Excel (**1,023/1,023** rows).
- Carry the desk's sales book: funded **\$4M against a \$2M monthly target** (200%) in each of April and May 2026; over quota every month since month two.

Founder & Software Engineer

Jan. 2024 – Present

SKomp Studio

Remote

- Founded a software studio and designed, shipped, and now operate its flagship client platform, **Solstice Pilates** (solsticepilates.ca), solo: a multi-tenant booking-and-payments system (**59-model Prisma schema**, **172 API routes**) with scheduling, FIFO waitlists and capacity gating, Square payments and recurring memberships with tax handling, digital waivers, and a 23-page admin portal.
- Runs live operations (**\$45K+ CAD processed** across **1,380 confirmed bookings**, 640 registered users) and ships a client-driven roadmap on **5,000+ automated tests**, with Playwright browser suites gating merges as required status checks, shadow-DB checks across every versioned migration, and Gitleaks secret scanning in CI.
- Engineered for money-handling correctness: idempotent Square payment processing, webhook signature verification, and row-level tenant isolation via a scoped Prisma client.
- Landed five production fixes in parallel in one agent-orchestrated session, each in its own git worktree with its own test-and-verify loop, merged through branch-protected PRs.
- Coached **100+ learners** through algorithms, systems design, and technical-interview prep in the studio's mentorship arm, taking each engagement from acquisition through delivery.

Junior Web Developer

May 2021 – Jun. 2022

Giftcash Inc.

Remote

- Migrated a legacy Python / Django monolith to Node.js on AWS Lambda, trading always-on server overhead for per-request billing, and tuned PostgreSQL with indexing, caching, and query rewrites that killed the sequential scans behind the gift-card lookup flow.
- Automated gift-card balance verification with a Puppeteer + Axios pipeline and stood up Jenkins and GitHub Actions CI/CD so merges deployed without manual release steps.

PROJECTS

Personal Agentic OS · multi-agent automation platform | *Python, TypeScript, Claude, systemd, Tailscale* 2026

- Operate the self-hosted multi-agent platform behind the document-validation service above: **30+ scheduled automations** and email-triggered workers that validate funding-deal packages in minutes, containing prompt-injection risk from untrusted attachments.
- Every untrusted worker runs in a kernel-namespaces sandbox (bubblewrap) behind fail-closed gates; secrets live in a 1Password vault, backups run nightly off-site, and a health watchdog emails on problem-state transitions.
- Runs an autonomous overnight pipeline that picks a candidate idea, builds and tests it in an isolated git worktree, and publishes behind a **12-check eval gate** (dependency vulnerabilities, secret hygiene, origin integrity, a model-based validator), refusal reasons persisted to an audit log; monitored from an iOS app over Tailscale; heavy jobs dispatch to a second compute node through a FastAPI work queue with artifact upload.
- Includes a fully local RAG stack (qwen2.5 on an RTX 3080) returning cited answers in **about 4s** at zero API cost, plus a Whisper voice-note router that files each note to the right project, tuned against Recall@10.

Mike Ross AI · legal RAG assistant (Delta Hacks 12) | *LangGraph, Gemini 2.0, MongoDB Atlas, FastAPI* 2026

- Built a legal assistant that answers Canadian-law questions with schema-enforced, quote-grounded citations: a LangGraph state machine routes intent and runs jurisdiction-aware vector retrieval (provincial queries auto-merge federal law) over a **16,000-chunk** corpus of seven Canadian statutes on MongoDB Atlas.
- Wired Gemini 2.0 Flash structured outputs to a FastAPI backend with multi-turn thread memory and a draft-to-PDF pipeline for legal notices; live at mike-ross.ca.

LLM Study Companion · MEng research system | *FastAPI, Next.js, pgvector* 2026

- Building the system around the research: event-driven hexagonal services (FastAPI, Next.js, Postgres + pgvector), AssemblyAI lecture transcription, and Cohere reranking, with the TLA+/SMT2-specified citation gate as the publish oracle for generated study notes.

Applify AI · resume-tailoring SaaS | *Next.js, Inngest, OpenAI* 2025 – Present

- Production SaaS that tailors a resume to a job posting; seven durable Inngest job types drive the OpenAI Responses API with schema-validated outputs, including a **self-correcting page-fit loop** (up to 20 checkpointed iterations of bullet reduction, LaTeX recompile, page-count check, about 65 durable steps per document); Stripe billing with idempotent webhooks; 98-test CI suite.

suleyman.io · personal site built as a product | *Next.js 16, Prisma, Neon, Vercel* 2026

- The site is a command-center HUD with a working in-browser terminal (custom REPL, commands and games), live Spotify visualizations, a GitHub-language constellation fed by the GitHub API, and a database-backed resume pipeline serving versioned resume variants from Postgres.
- The resume pipeline behind this document runs on it: server-seeded Prisma rows, versioned LaTeX variants, and ATS-audited PDF builds.

SKomp Forge · agentic iOS training coach | *Swift 6, SwiftUI, SwiftData, OpenAI Realtime, HealthKit* 2025 – Present

- Agentic training coach across 11 Swift 6 modules with **52 decision records** carrying the tradeoffs; its agent exposes **18 tool schemas** pinned in Swift and Python and diff-tested so the two can never drift.
- Full-duplex voice on **gpt-realtime** with server-side VAD and barge-in, holding **P90 313ms** to first audio against a 1,500ms gate; the phone never holds an API key, only a 60-second ephemeral token.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, Java, C, Swift

Financial & Domain: Equipment & lease finance, amortization math (PMT/PV/FV/RATE/IRR, TValue-validated), manufacturer subsidy & blended-rate modeling, credit-scorecard & recovery modeling, deal-funding workflows

Backend & Web: Next.js (App Router, Server Actions), React, Node.js, FastAPI, Prisma, Tailwind, PostgreSQL (Neon), Redis, Vercel, REST APIs

AI / Agentic: Agent orchestration, LLM eval pipelines, RAG + reranking, local inference (Ollama, qwen2.5), Whisper, prompt-injection mitigation

Payments & Integrations: Square (orders, payments, webhooks), Meta CAPI, Google Ads server-side conversions, Gmail / Workspace API

Practices: TDD (Vitest, Playwright), CI/CD, branch-protected PRs, systemd automation, multi-tenant SaaS